

The Perceptron to the MLP

Scaler School of Technology | Deep Learning I | Week 2 — Session 1/3 | 6 May 2026



What Is This Lecture About?

We'll build neural networks from the ground up — starting with a single neuron (which is just logistic regression with a new name), understanding why activation functions matter, stacking neurons into multi-layer perceptrons, and solving the classic XOR problem that nearly killed the field.

1 Single Neuron & Activation Functions

How a neuron computes: weighted sum + bias + activation. Why sigmoid, tanh, and ReLU exist and when to use each.

2 Multi-Layer Perceptrons & Forward Pass

Stacking layers to learn complex decision boundaries. How data flows through a network: multiply, add, activate, repeat.

3 The XOR Problem

Why a single neuron can't solve XOR, and how adding one hidden layer fixes it. The historical moment that nearly ended neural networks.

What to Expect in the Session

- Hands-on coding: build a neuron and an MLP in PyTorch
- Visual demos of activation functions and decision boundaries
- 2 short quizzes + a final 3-question quiz
- The XOR case study — from failure to breakthrough
- Step-by-step forward pass calculation
- Live drawing of network architectures on iPad

How to Prepare (Optional but Helpful)

- Review logistic regression from Classical ML (sigmoid, decision boundaries)
- Recall what a linear decision boundary looks like
- Think about: what happens if you combine multiple linear classifiers?
- Optional: try the XOR truth table ($0 \oplus 0=0$, $0 \oplus 1=1$, $1 \oplus 0=1$, $1 \oplus 1=0$)
- Ensure Google Colab access with GPU runtime
- Have Session 1 concepts fresh: features, representations, UAT

Duration: ~2 hours

Prerequisites: Session 1, Classical ML

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